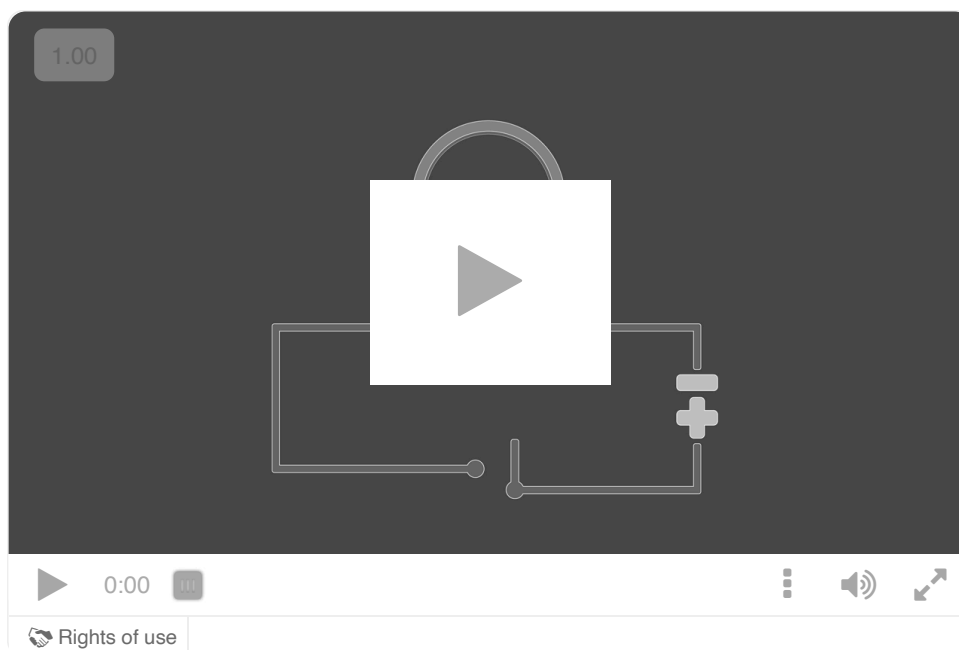


Electrodes in electrochemical cell

Learning outcomes

By the end of this section you should be able to label electrodes as anode and cathode and identify their polarities in voltaic and electrolytic cells, based on the type of reaction occurring at the electrode.

Batteries are used to power devices, like in the example of the circuit in **Video 1** where a battery is lighting a lamp.



Video 1. The Battery Supplies Energy to the Lamp.

 More information for video 1

A video illustrates a simple electrical circuit with a light bulb, a battery, and a switch. When the circuit is open, the bulb remains off. As the switch is closed, completing the circuit, the bulb lights up, indicating the flow of electricity. This continues in a loop as the video progresses.

An animation video illustrates a simple electrical circuit with a light bulb, a battery, and a switch. When the circuit is open, the bulb remains off. As the switch is closed, completing the circuit, the bulb lights up, indicating the flow of electricity. This continues in a loop as the animation progresses.

In **Video 1** the battery is represented by the positive and negative signs. Have you noticed when using a battery that there are different ends? Batteries need to be inserted in the correct orientation in order to work. Why do you think this is?

Batteries are used to supply electricity; the movement of electrons through a conducting material. What does **Video 1** tell you about the 'flow' of electrons? Electrons flow in one direction, like water flowing in a pipe. How might that relate to the orientation of a battery?

Electrochemical cells

Redox reactions happen in systems that are called electrochemical cells. The term 'cell' was used for the first time by an Italian physicist Alessandro Volta in 1800 when he invented the first electrochemical cell. He named his battery a 'voltaic pile'. Perhaps you can see why in **Figure 1**?

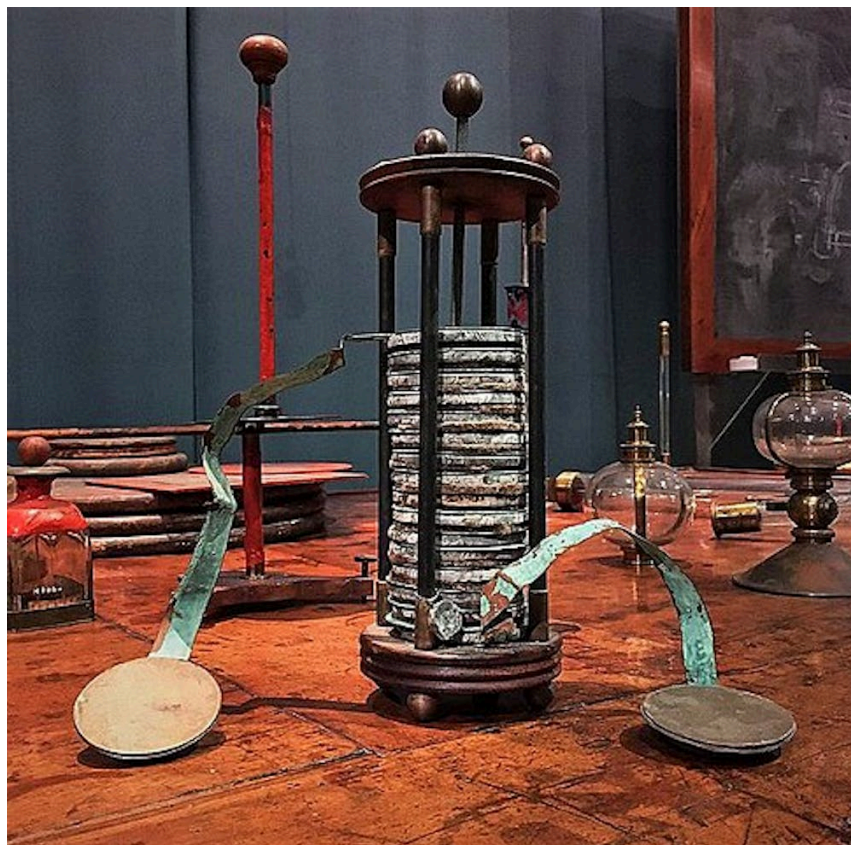


Figure 1. A voltaic pile, the first electrochemical cell.

Source: "Pila di volta

(https://commons.wikimedia.org/wiki/File:Pila_di_volta.jpg)" by Markkv is

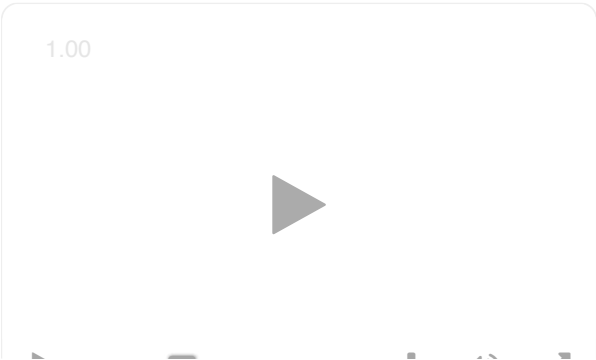
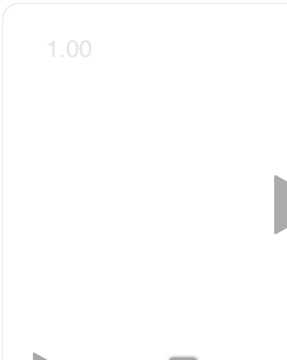
licensed under CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/deed.en>)

Because there is a transfer of electrons due to chemical reactions happening inside cells, they are called electrochemical cells. For the cells to work, the redox reaction happening needs to be split into the two half-reactions, but they need to be connected by a conducting wire.

There are two types of electrochemical cells, and they are called voltaic cells and electrolytic cells. Both involve conversions between chemical and electrical energy but in opposite directions. Check **Table 1** for a comparison between the two types

of cells.

Table 1. A comparison between voltaic (also called galvanic cells) and electrolytic cells.

Voltaic cells	Electrolytic cells
Convert chemical energy to electrical energy	Convert electrical energy to chemical energy
Energy from spontaneous exothermic chemical reactions is converted to electrical energy	Energy is provided by an external source to drive a non-spontaneous chemical reaction
Energy is released, as you learned in <u>subtopic R2.1</u> (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43479/).  Interactive 1. Voltaic Cell.  More information for interactive 1 The animation video demonstrates the concept of voltaic cells using the movement of a ball. The ball initially moves along a small horizontal path and then rolls downhill, representing the release of energy. After that, the ball continues in a horizontal direction, forming a stepped-line pattern. This video highlights how a spontaneous reaction occurs without any external force and how a voltaic cell naturally converts chemical energy into electrical energy without external input.	Energy is required, as you learned in <u>subtopic R2.1</u> (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43479/).  Interactive 2. Electrolytic Cell.  More information for interactive 2 The animation video illustrates the concept of electrolytic cells using the movement of a ball. The ball initially moves in a small horizontal path and then moves uphill, indicating that energy is being supplied. After reaching the top, the ball continues in a horizontal direction, forming a stepped-line pattern. This video highlights how a non-spontaneous reaction requires an external energy source to occur and how an electrolytic cell converts electrical energy into chemical energy.

For either of the cells to work, two electrodes are required, each containing one of the half-reactions. Electrodes are conductors of electricity used to make contact with a non-metallic part of a circuit, such as a solution in a cell, called the electrolyte. Generally, the electrodes are metal, but they can also be other materials that conduct electricity, such as graphite.

In **Figure 2**, you can see a general diagram of a voltaic and an electrolytic cell for comparison.

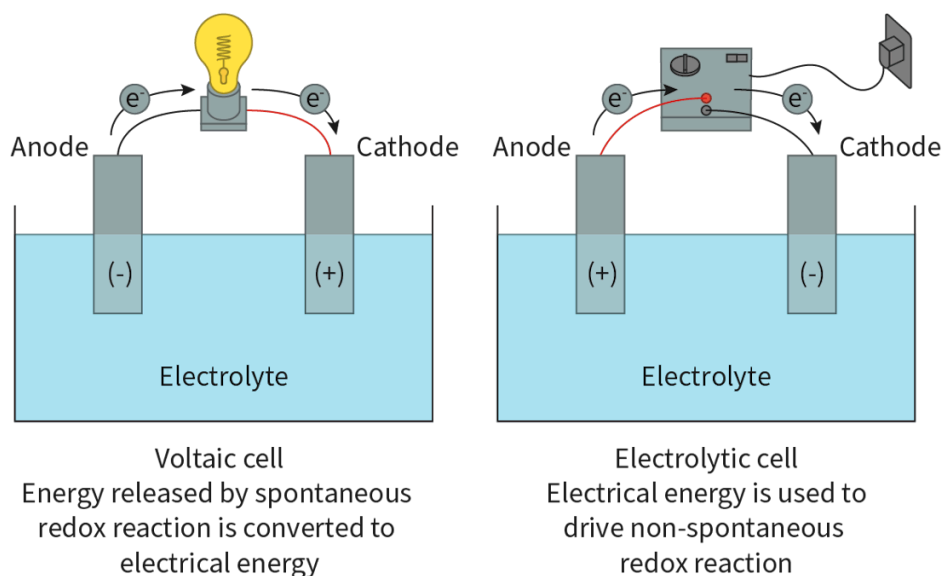


Figure 2. A voltaic and an electrolytic cell for comparison.

More information for figure 2

The image is a diagram comparing a voltaic cell and an electrolytic cell. On the left side, the voltaic cell is shown with one anode and one cathode submerged in an electrolyte solution. The anode is marked with a negative sign (-), while the cathode is marked with a positive sign (+). Above the electrolyte, there is a wire connecting the two electrodes with a light bulb, indicating the flow of electrons from the anode to the cathode, generating electrical energy.

On the right side, the electrolytic cell is similarly structured with an anode and a cathode, marked with a positive sign (+) and a negative sign (-) respectively. Here, the flow of electrons is driven by an external power source displayed between the electrodes, emphasizing that electrical energy is used to drive a non-spontaneous redox reaction.

Text beneath the voltaic cell reads: "Voltaic cell: Energy released by spontaneous redox reaction is converted to electrical energy." Beneath the electrolytic cell, the text reads: "Electrolytic cell: Electrical energy is used to drive non-spontaneous redox reaction."

[Generated by AI]

As seen in **Figure 2**, electrochemical cells contains two electrodes:

- An anode: where oxidation takes place and electrons are lost by the species being oxidised.
- A cathode: where reduction occurs and electrons are gained by the species being reduced.

You can come up with your own mnemonic to remember this information, but here is one to get you started:

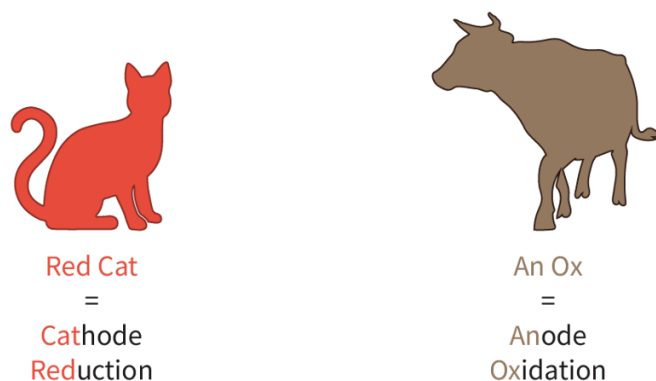


Figure 3. A mnemonic to remember what happens at each electrode.

You may have noticed in **Figure 2**, that depending on the electrochemical cell, the electrodes will have different polarities. This polarity refers to the relative charge of the two electrodes in the cell – where electrons are being released, there is a more negative charge (negative pole). In voltaic cells, the electrons released are dependent on the species undergoing oxidation. In electrolytic cells, electrons are initially provided by the battery and they enter the cell at the cathode. However, note that electrons always flow from the anode to the cathode in both types of cells. This is why you need to place batteries with the correct orientation in a device in order for it to work.

Table 2. A summary of the polarity of electrodes on electrochemical cells.

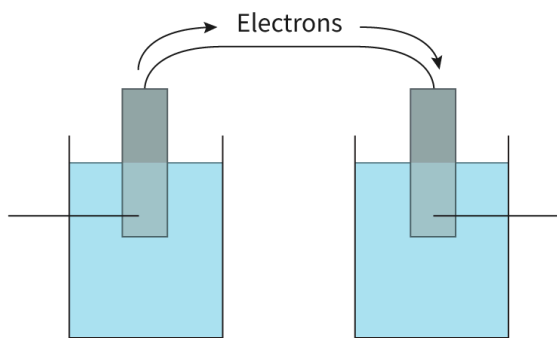
Electrode	Reaction happening	Voltaic cells	Electrolytic cell
Anode	Oxidation	Negative	Positive
Cathode	Reduction	Positive	Negative

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Communication skills — Using terminology, symbols
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/pair activity

Use **Interactive 3** to correctly position the labels on the diagrams. Click on the arrow at the bottom right corner for the second part the interactive.

Voltaic cells – where energy is produced

☒ Check

Interactive 3. A Voltaic and an Electrolytic Cell for Comparison.